

Geocomputation & Machine Learning course

Quick recap

This was a lecture session on Bash scripting and GDAL commands for geospatial data processing. Giuseppe led the class through reviewing previous assignments using AWK for data manipulation, where students like Abdul shared their experiences working with time series data and flood extent maps. The session covered the difference between using Bash for simple file operations versus AWK for deeper data processing within columns, with examples of joining multiple files based on unique IDs and creating efficient data extraction workflows. The second part focused on introducing GDAL commands for raster and vector data manipulation, including reading metadata with `gdalinfo`, understanding data types and no-data values, performing raster calculations, and resampling techniques. Students discussed their experience with various file formats including HDF, NetCDF, and GeoTIFF, and the importance of proper projection settings and pixel resolution considerations. The lecture concluded with practical examples of geolocation extraction and resampling operations, with students like Ali and Sebastián asking questions about reprojection methods and installation procedures for their Ubuntu systems.

Next steps

Abdul

- Complete the next assignment involving GDAL and other tools to find the maximum depth using TIFF files, focusing on the bridge downstream to check at what discharge the water goes out of the bridge, within the next two days.

Ali

- Resolve the virtual machine issue by installing and configuring the required environment directly on Ubuntu 26.04, following Giuseppe's instructions, and clone the exercise repository into the appropriate directory.

Giuseppe

- Record the next two sessions for attendees who cannot attend, as requested by Sebastián.

Collaboration

- All attendees: For the next class, prepare a table using GDAL location info, where each row contains the latitude, longitude, and corresponding raster values for multiple points, formatted as a single line per point.

Summary

AWK and Bash Commands Discussion

Giuseppe led a discussion on the experience with Bosch and AWK commands, where Abdul shared his work on time series data analysis using AWK to extract and process information, finding it more efficient than manual Excel operations. Andrea expressed interest in understanding when to use Bash versus AWK, to which Giuseppe explained that Bash is suitable for simple file operations like searching or extracting lines, while AWK is needed for more complex data manipulations within columns. Giuseppe also reviewed the syllabus and provided guidance on accessing and using .sh files for bash scripting, suggesting editors like gedit or Emacs for further work.

Emacs and Data Processing Demo

Giuseppe demonstrated how to open Emacs directly in the terminal, explaining its advantages for working with HPC systems due to reduced delay in network communication. He then walked through a script example that processes GCM data to create a table with ID, latitude, longitude, and mean values, showing two different approaches: a combination of Bash and AWK commands, and a looping method using AWK variables. Giuseppe explained the join command for combining files based on unique IDs and emphasized the importance of sorting files before joining, particularly for large datasets.

AWK Data Processing Demonstration

Giuseppe demonstrated how to use AWK to process and organize data files, showing two approaches: a loop-based method and a more efficient single-pass method using AWK commands. He explained that AWK is effective for well-organized data but becomes challenging with messy data, suggesting Python or R as alternatives for complex datasets. Abdul mentioned having many text files from different regions of Italy that need processing, and Giuseppe advised that Python would be more suitable for handling data from different sources due to potential structural differences.

CSV and GDAL Data Handling

Giuseppe discussed CSV data handling and field separators in Bash, explaining the importance of using the correct field separator identifier for CSV files. He then introduced the second part of the lecture focused on GDAL commands for raster and vector manipulation in the Geo environment. Giuseppe explained GDAL's history, advantages, and recent updates, including the new GDAL application with different command syntax. He also covered various raster and vector formats supported by GDAL, noting that GeoTIFF is the default output format.

Data Formats in Hydraulic Analysis

Giuseppe led a discussion on data formats and projections used in hydraulic analysis and geospatial data processing. Participants, including Abdul, Areeba, and Eva, shared their experiences with formats such as HEC-RAS HDF files, NetCDF, and data cubes. Giuseppe explained the use of GDAL for supporting various vector and raster formats and introduced a resource for finding EPSG codes and projection information. He emphasized the importance of using the EPSG website to verify projection strings and provided guidance on accessing different projection file formats like WKT, JSON, and PRJ.

JupyterLab Raster File Operations

Giuseppe demonstrated how to use JupyterLab to work with raster files, explaining the importance of checking metadata before opening large files. He showed how to use commands like ``gdalinfo`` to retrieve information about file size, pixel resolution, projection, and no-data values. The discussion included an explanation of how pixel sizes are calculated in degrees and their relationship to meters, with examples of different resolutions like 0.5 degrees and 30-second intervals. Giuseppe also mentioned the potential need to resample datasets when pixel sizes differ between X and Y coordinates.

Data Types in Geographic Data

Giuseppe explained the concept of data types in computer science, particularly focusing on how different data types affect storage size and precision when working with geographic data like temperature and precipitation. He demonstrated how byte, unsigned, integer, and floating-point data types have specific ranges and explained that choosing the wrong data type can lead to data loss or excessive file sizes. Giuseppe clarified that when working with GDAL, data types can be specified, but if not specified, the output will match the input data type, emphasizing the importance of understanding data ranges when converting between different types.

Data Reference and Conversion Guidelines

Giuseppe instructed the team to always reference a specific table for data information and emphasized the importance of checking it periodically, especially for larger numbers like 4 million. He discussed data types, particularly focusing on temperature data and the concept of scale and offset in datasets. Giuseppe explained how to adjust data values by adding or subtracting 50 to convert them into bytes and save them with appropriate offsets or scales. He mentioned plans to search for a file containing offset and scale information during the break and encouraged team members to check their own data types and JITA information in their datasets.

Handling No Data in Rasters

Giuseppe explained the importance of understanding no data values in raster datasets, emphasizing that no data is actually a specific number rather than nothing. He demonstrated how to use GDAL commands to check minimum and maximum values, including handling no data values correctly. Ali asked about extracting values from raster layers for modeling purposes, particularly for flood prediction, and Giuseppe explained that points falling on no data areas would return -999 values, suggesting either removing those points or using interpolation techniques to handle the gaps.

Raster Data Processing Techniques

Giuseppe explained the importance of correctly labeling no-data values in raster images, using minus 9999 as an example for DTM data. He demonstrated how to use GDAL to compute minimum and maximum values across multiple files and emphasized the importance of

checking data types, particularly when performing floating-point operations like NDVI calculations. Giuseppe also introduced resampling techniques for changing pixel resolution, though the transcript ended before he could complete the explanation of resampling.

GDAL Resampling Techniques Training

Giuseppe taught the class about resampling techniques in GDAL, explaining different methods like nearest neighbor, bilinear, and cubic interpolation. He demonstrated how to use GDAL commands for resampling, reprojection, and extracting geolocation information from raster data. The class discussed practical applications, with Ali sharing his work on resampling 72 layers for North Atlantic Canada, and Annabell asking about using GDAL commands versus R functions for reprojection. Giuseppe clarified that the bash commands shown are actually the same underlying method used in R and Python, just implemented directly in GDAL.